CUBE.AI: An Agentic AI Platform for Automated Radiology Research

Team 9: Ahmed Soliman, Ibrahim Fateen, Shahd Ali, Meram Mahmoud, Mostafa Ali
Supervised By:
Meena M. Makary, PhD – Hossam El-Rewaidy, PhD
Systems and Biomedical Engineering
Cairo University

I. Problem Definition

Radiology research has become a cornerstone of modern clinical workflows, playing a central role in diagnostic tool development, biomarker discovery, and therapeutic decision-making [1], [2]. Despite its significance, producing a radiology study remains a protracted and labor-intensive undertaking [3]. The research pipeline spans hypothesis formulation, study design, imaging data acquisition, region-of-interest (ROI) segmentation, quantitative feature extraction, predictive modeling, and scientific reporting [4]. each stage relying on a distinct set of specialized software tools . This fragmentation forces researchers to manually transfer data between environments, convert between incompatible formats, and maintain proficiency in computer science, statistics, and data engineering - fields that are outside their clinical expertise [5] [6]. The resulting technical barrier limits the pace of discovery and prevents many radiologists from fully realizing their research potential [3].

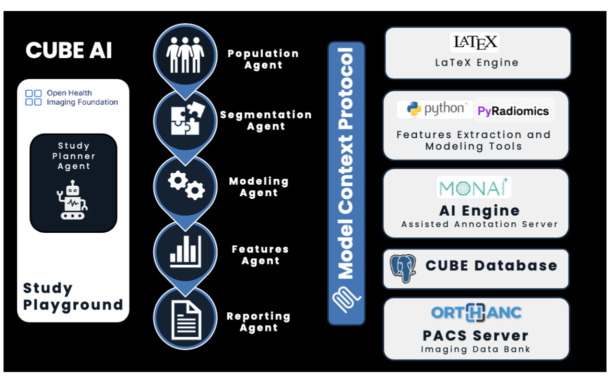

Fig. 1: CUBE unified agentic architecture.

II. Proposed Solution

CUBE is an agentic AI platform designed to automate and orchestrate the entire radiology research lifecycle within a single, cohesive environment [7]. Receiving only a natural-language research hypothesis as input, CUBE first uses a planner agent to structure the hypothesis according to the PICO (Population, Intervention, Comparison, Outcome) framework, and generate a comprehensive study plan to fully explore the hypothesis. It assigns a specialized agent to each phase of the plan, and writes a detailed job description for each agent. Through natural language interaction, the user can change some aspects of the plan, and once they're satisfied, they can start the automated execution. Segmentation is performed via AI-driven models built on the MONAI framework [8], features are extracted using established radiomic pipelines [9], and downstream machine learning and deep learning models are trained and validated automatically [10], [11]. Results are compiled into publication-ready documents via an integrated LaTeX engine, eliminating the need for any tool switching throughout the process.

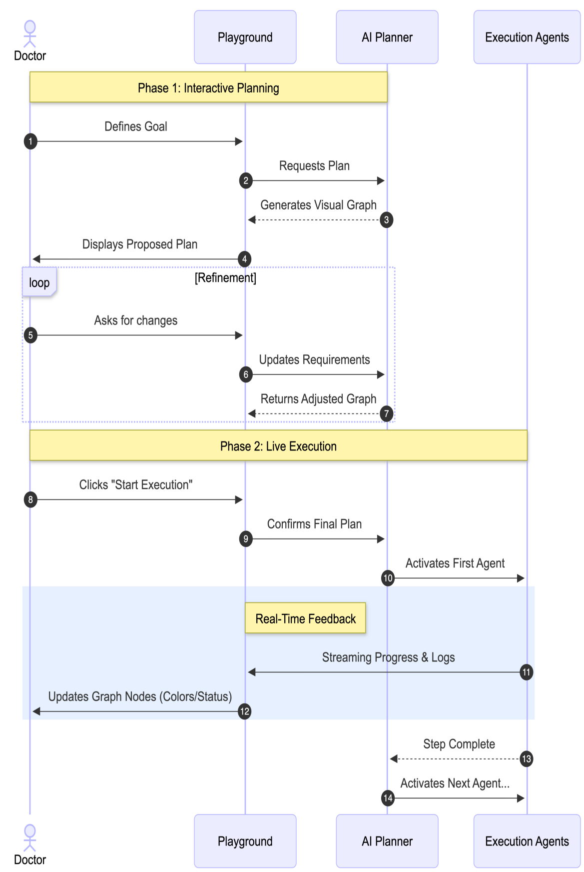

Fig. 2: CUBE Sequence Diagram.

III. Bridging the Gap

CUBE bridges the gap between complex research methodology and clinical accessibility through three key mechanisms. First, it integrates directly with hospital PACS infrastructure, enabling cohort construction and imaging retrieval without manual data export or format conversion. Second, it provides an interactive annotation layer combining AI-predicted segmentations with user-guided refinement tools so that ROI delineation remains both efficient and clinically supervised [8]. Third, a glass-box observability layer exposes the internal state of every agent at each execution step, allowing the radiologist to inspect intermediate decisions, intervene when necessary, and maintain full methodological control [12], [13]. This transparent collaboration model ensures that automation accelerates rather than supplants expert judgment.

IV. Conclusion

CUBE addresses the dual challenge of workflow fragmentation and technical inaccessibility that currently impedes radiology research. By consolidating the full research pipeline from literature review to manuscript preparation within an orchestrated, natural-language-driven environment, CUBE dramatically reduces time-to-insight while preserving the radiologist's role as the scientific authority [10], [14]. The platform democratizes quantitative imaging research, enabling clinicians with limited computational background to design, execute, and report reproducible studies independently. Future work will focus on augmenting CUBE with literature review and knowledge synthesis capabilities, allowing the system to ground study design and analysis in up-to-date scientific evidence.

References

[1] P. Lambin, E. Rios-Velazquez, R. Leijenaar, S. Carvalho, R. G. van Stiphout, P. Granton, C. M. Zegers, R. Gillies, R. Boellaard, A. Dekker, and H. J. Aerts, "Radiomics: Extracting more information from medical images using advanced feature analysis," *European Journal of Cancer*, vol. 48, no. 4, pp. 441–446, 2012.

[2] H. J. Aerts, E. R. Velazquez, R. T. Leijenaar, C. Parmar, P. Grossmann, S. Carvalho, J. Bussink, R. Monshouwer, B. Haibe-Kains, D. Rietveld, F. Hoebers, M. M. Rietbergen, C. R. Leemans, A. Dekker, J. Quackenbush, R. J. Gillies, and P. Lambin, "Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach," *Nature Communications*, vol. 5, p. 4006, 2014.

[3] K. Chockley and E. Emanuel, "The end of radiology? three threats to the future practice of radiology," *Journal of the American College of Radiology*, vol. 13, no. 12, Part A, pp. 1415–1420, 2016. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1546144016305907

[4] A. S. Doria, G. Tomlinson, J. Beyene, and R. Moineddin, Eds., *Research Methods in Radiology: A Practical Guide*. Stuttgart: Georg Thieme Verlag KG, 2018.

[5] M. A. Morris, B. Saboury, B. Burkett, J. Gao, and E. L. Siegel, "Reinventing radiology: Big data and the future of medical imaging," *Journal of Thoracic Imaging*, vol. 33, no. 1, pp. 4–16, 2018.

[6] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, no. 1, pp. 44–56, 2019.

[7] C. Bluethgen, D. V. Veen, D. Truhn, J. N. Kather, M. Moor, M. Polacin, A. Chaudhari, T. Frauenfelder, C. P. Langlotz, M. Krauthammer, and F. Nooralahzadeh, "Agentic systems in radiology: Design, applications, evaluation, and challenges," 2025. [Online]. Available: https://arxiv.org/abs/2510.09404

[8] M. J. Cardoso, W. Li, R. Brown, N. Ma, E. Kerfoot, Y. Wang, B. Murrey, A. Myronenko, C. Zhao, D. Yang, V. Nath, Y. He, Z. Xu, A. Hatamizadeh, W. Zhu, Y. Liu, M. Zheng, Y. Tang, I. Yang, M. Zhu, L. Bi, S. Bengio, R. Bhatt, D. Lachinov, T. Han, F. Liebel, F. Perez-Garcia, N. Savill, D. Clunie, W. Huang, Q. Dou, A. Feng, H.-P. Meinzer, D. Xu, S. Bisdas, A. Flanders, G. Feng, M. Xu, P. Zwetsloot, S. Giannarou, M. Alber, M.-M. Rohe, and J. a. Cardoso, "MONAI: An open-source framework for deep learning in healthcare," *arXiv preprint arXiv:2211.02701*, 2022.

[9] J. J. van Griethuysen, A. Fedorov, C. Parmar, A. Hosny, N. Aucoin, V. Narayan, R. G. Beets-Tan, J.-C. Fillion-Robin, S. Pieper, and H. J. Aerts, "Computational radiomics system to decode the radiographic phenotype," *Cancer Research*, vol. 77, no. 21, pp. e104–e107, 2017.

[10] D. Shen, G. Wu, and H.-I. Suk, "Deep learning in medical image analysis," *Annual Review of Biomedical Engineering*, vol. 19, pp. 221–248, 2017.

[11] E. Tzanis and M. E. Klontzas, "maistro: An open-source multi-agent system for automated end-to-end development of radiomics and deep learning models for medical imaging," *European Journal of Radiology Artificial Intelligence*, vol. 4, p. 100044, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S3050577125000428

[12] M. T. Contaldo *et al.*, "Ai in radiology: Navigating medical responsibility," *Diagnostics*, vol. 14, no. 14, p. 1506, 2024.

[13] R. Lawrence, E. Dodsworth, E. Massou, C. Sherlaw-Johnson, A. I. Ramsay, H. Walton *et al.*, "Artificial intelligence for diagnostics in radiology practice: a rapid systematic scoping review," *eClinicalMedicine*, p. 103228, 2025.

[14] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "AI in health and medicine," *Nature Medicine*, vol. 28, no. 1, pp. 31–38, 2022.